

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Distance travelled / length of track / 2.5 m Accept: [same] car / [same] track	1			1		1
	(b)	(i)		Mean speed = $\frac{2.5}{4.0}$ (1-substitution) = 0.625 (1) [m/s] accept 0.63 and 0.6 but not 0.62 [m/s] Answer of 0.62 on the answer line with no workings shown award 1 mark	1	1		2	2	2
		(ii)	I.	The [mean] time or it decreases (1) at a decreasing rate / at smaller intervals (1) N.B. treat any reference to numbers as neutral		2		2	2	2
			II.	Increases accept quicker or faster		1		1	1	1
	(c)	(i)		It [0.2 s] is not the smallest range / 0.1 s is the smallest range (1) <u>Most repeatable data is for 10 cm</u> (1)			2	2	2	2
		(ii)		Reference to human error or reaction times (1) so the timing is closer to the true value / more accurate (1) N.B. treat any reference to precision or reliability as neutral			2	2		2
				Question 4 total	2	4	4	10	7	10